

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (BSc IT Semester V)

Enterprise Java Practical

Module - I

Practical – VI

Roll No.: T007	Name :Subhashree jena
Class: TYIT	Batch:1
Date of Assignment: 22/08/26	Date/Time of Submission: 22/08/26

6. Implement the following Intermediate JSP Applications.

a. Develop a JSP application to demonstrate the use of JSP Directives and Implicit Objects.

CODE:-

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.Date" %>
<html>
<head>
<title>JSP Directives and Implicit Objects</title>
</head>
<body>
<h2>JSP Directives and Implicit Objects</h2>
<h3>JSP Directives</h3>
<p>Language: Java</p>
<p>Content Type: <%= response.getContentType() %></p>
<p>Current Date: <%= new Date() %></p>
<h3>Implicit Objects</h3>
<p>1. Request Method: <%= request.getMethod() %></p>
<p>2. Response Content Type: <%= response.getContentType() %></p>
<p>3. Session ID: <%= session.getId() %></p>
<p>4. Application Server: <%= application.getServerInfo() %></p>
<p>5. Page Name: <%= page.getClass().getName() %></p>
<p>6. PageContext: <%= pageContext.getClass().getName() %></p>
<p>7. Request Parameter: <%= request.getParameter("name") %></p>
<p>8. Config Servlet Name: <%= config.getServletName() %></p>
<p>9. Exception Object: Exception object is available only in error pages.</p>
</body>
</html>
```

OUTPUT:-

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %> <%@ page import="java.util.Date" %>
```

JSP Directives and Implicit Objects

JSP Directives

Language: Java

Content Type: <%= response.getContentType() %>

Current Date: <%= new Date() %>

Implicit Objects

1. Request Method: <%= request.getMethod() %>

2. Response Content Type: <%= response.getContentType() %>

3. Session ID: <%= session.getId() %>

4. Application Server: <%= application.getServerInfo() %>

5. Page Name: <%= page.getClass().getName() %>

6. PageContext: <%= pageContext.getClass().getName() %>

7. Request Parameter: <%= request.getParameter("name") %>

8. Config Servlet Name: <%= config.getServletName() %>

9. Exception Object: Exception object is available only in error pages.

b. Develop a JSP application to demonstrate Form Processing and Input Validation.

CODE:-

Admission.jsp:

```
<html>
<head>
  <title>Student Admission Form</title>
</head>
<body>
  <h2>Student Admission Form</h2>
  <form action="process.jsp" method="post">
    Name:
    <input type="text" name="name">
    <br><br>

    Age:
    <input type="text" name="age">
    <br><br>

    Email:
    <input type="text" name="email">
    <br><br>

    Mobile:
    <input type="text" name="mobile">
    <br><br>

    Gender:
    <input type="radio" name="gender" value="Male"> Male
    <input type="radio" name="gender" value="Female"> Female
    <br><br>

    Course:
    <select name="course">
      <option value="">Select Course</option>
      <option value="BSc IT">BSc IT</option>
      <option value="BCA">BCA</option>
      <option value="MSc IT">MSc IT</option>
      <option value="MCA">MCA</option>
    </select>
    <br><br>

    Hobbies:
    <input type="checkbox" name="hobby" value="Reading"> Reading
    <input type="checkbox" name="hobby" value="Sports"> Sports
    <input type="checkbox" name="hobby" value="Music"> Music
    <br><br>

    Address:
    <textarea name="address"></textarea>
    <br><br>
    <input type="submit" value="Submit">
  </form>
</body>
</html>
```

Process.jsp:

```
<html>
<head>
  <title>Admission Details</title>
</head>
<body>
  <h2>Admission Details</h2>
  <%
    String name = request.getParameter("name");
    String age = request.getParameter("age");
    String email = request.getParameter("email");
    String mobile = request.getParameter("mobile");
    String gender = request.getParameter("gender");
    String course = request.getParameter("course");
```

```

String address = request.getParameter("address");
String[] hobbies = request.getParameterValues("hobby");
boolean valid = true;
if (name == null || name.trim().equals("")) {
    out.println("Name is required.<br>");
    valid = false;
}
if (age == null || age.trim().equals("")) {
    out.println("Age is required.<br>");
    valid = false;
}
if (email == null || email.trim().equals("")) {
    out.println("Email is required.<br>");
    valid = false;
}
if (mobile == null || mobile.trim().equals("")) {
    out.println("Mobile number is required.<br>");
    valid = false;
}
if (gender == null) {
    out.println("Please select gender.<br>");
    valid = false;
}
if (course == null || course.equals("")) {
    out.println("Please select a course.<br>");
    valid = false;
}
if (hobbies == null) {
    out.println("Please select at least one hobby.<br>");
    valid = false; }
if (address == null || address.trim().equals("")) {
    out.println("Address is required.<br>");
    valid = false; }
if (valid) {
    %>
    <h3>Student Admission Successful</h3>
    Name: <%= name%><br><br>
    Age: <%= age%><br><br>
    Email: <%= email%><br><br>
    Mobile: <%= mobile%><br><br>
    Gender: <%= gender%><br><br>
    Course: <%= course%><br><br>
    Hobbies:
    <%
        for (String hobby : hobbies) {
            out.println(hobby + " ");
        }
    %>
    <br><br>
    Address: <%= address%><br><br>
    <%
    }
    %>
    </body>
</html>

```

Output:

Admission Details

Student Admission Successful

Name: Subhashree Jena

Age: 20

Email: subhjenamvlu@gmail.com

Mobile: 989241567

Gender: Female

Course: BSc IT

Hobbies: Music

Address: Andheri E

Student Admission Form

Name:

Age:

Email:

Mobile:

Gender: ☐ Male ☒ Female

Course:

Hobbies: ☐ Reading ☐ Sports ☒ Music

Address:

c. Develop a JSP application to display data from a MySQL Database using JDBC.

CODE:-

SQL CODE:

```
CREATE DATABASE ecommerce;
USE ecommerce;
Database changed
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    brand VARCHAR(50),
    price DECIMAL(10,2),
    quantity INT,
    rating DECIMAL(2,1),
    discount DECIMAL(5,2),
    description VARCHAR(255)
);
```

Index.jsp:

```
<%@ page language="java"
    contentType="text/html; charset=UTF-8"
    import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>E-Commerce Products</title>
<style>
    body {
        font-family: Arial, sans-serif;
        background-color: #f2f2f2;
        margin: 30px;
    }
    h2 {
        text-align: center;
        color: #333;
    }
    tr:nth-child(even) {
        background-color: #f2f2f2;
    }
    tr:hover {
        background-color: #D6EAF8;
    }
}
```

```

</style>
</head>
<body>
<h2>E-Commerce Product List</h2>
<table>
<tr>
<th>Product ID</th>
<th>Product Name</th>
<th>Category</th>
<th>Brand</th>
<th>Price</th>
<th>Quantity</th>
<th>Rating</th>
<th>Discount</th>
<th>Description</th>
</tr>
<%
Connection con = null;
Statement stmt = null;
ResultSet rs = null;
try {
Class.forName("com.mysql.jdbc.Driver");
con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/ecommerce",
"root",
"root"
);
stmt = con.createStatement();
rs = stmt.executeQuery("SELECT * FROM product");
while (rs.next()) {
%>
<tr>
<td><%= rs.getInt("product_id")%></td>
<td><%= rs.getString("product_name")%></td>
<td><%= rs.getString("category")%></td>
<td><%= rs.getString("brand")%></td>
<td><%= rs.getDouble("price")%></td>
}
}
%>
</table>
</body>
</html>

```

OUTPUT:-

```
mysql> SELECT * FROM product;
```

product_id	product_name	category	brand	price	quantity	rating	discount	description
1	iPhone 15	Mobile	Apple	69999.00	10	4.8	10.00	Apple iPhone with advanced camera and performance
2	Galaxy S24	Mobile	Samsung	64999.00	15	4.7	12.00	Samsung smartphone with AMOLED display
3	MacBook Air M2	Laptop	Apple	89999.00	8	4.9	8.00	Lightweight laptop powered by Apple M2 chip
4	WH-1000XM5	Headphones	Sony	29999.00	20	4.6	15.00	Wireless noise cancelling headphones
5	Inspiron 15	Laptop	Dell	55999.00	12	4.5	10.00	Dell laptop suitable for everyday computing

5 rows in set (0.00 sec)

E-Commerce Product List								
Product ID	Product Name	Category	Brand	Price	Quantity	Rating	Discount	Description
1	iPhone 15	Mobile	Apple	₹69999.0	10	4.8 ⭐	10.0%	Apple iPhone with advanced camera and performance
2	Galaxy S24	Mobile	Samsung	₹64999.0	15	4.7 ⭐	12.0%	Samsung smartphone with AMOLED display
3	MacBook Air M2	Laptop	Apple	₹89999.0	8	4.9 ⭐	8.0%	Lightweight laptop powered by Apple M2 chip
4	WH-1000XM5	Headphones	Sony	₹29999.0	20	4.6 ⭐	15.0%	Wireless noise cancelling headphones
5	Inspiron 15	Laptop	Dell	₹55999.0	12	4.5 ⭐	10.0%	Dell laptop suitable for everyday computing